TechPad Schematic Guide

Oct 17, 2026 · @MoonRock

Quick answers

You connect 13 switches: the 12 key switches plus the encoder's push-click. Each one gets its own 1N4148 diode, so the matrix holds 13 diodes too.

This guide covers the schematic only. Part 1 explains how schematics work, part 2 explains the circuits on TechPad, and part 3 walks you through building it in KiCad from where your file stands now. Footprints and the PCB come after the schematic passes review.

Part	Qty	KiCad symbol
Seeed XIAO RP2040	1	XIAO_RP2040:MOUDLE-SEEEDUINO-XIAO (care package)
MX key switch	12	Switch:SW_Push (your SW_Push_45deg also works)
1N4148 diode	13	Diode:1N4148 (your Device:D also works)
EC11 encoder with push	1	Device:RotaryEncoder_Switch
0.91 inch OLED, 4-pin I2C	1	Connector_Generic:Conn_01x04
SK6812 MINI-E RGB LED	12	SK6812MINI-E:SK6812MINI-E (care package)
Mounting hole	4	Mechanical:MountingHole

Where your file stands (read 19:15 today): 9 keys and 9 diodes wired as a plain 3x3 matrix with labels R1 to R3 and C1 to C3. There are 2 LEDs, a +5V symbol, and 4 PWR_FLAGs (3 of them on GND). The OLED connector is placed but has no wires. **No wire or label reaches the XIAO yet**, so the board would do nothing as drawn. The library tables still use B:/ paths.

Crash course 1: how a schematic works

A schematic is a list of which pins connect to which, drawn as a picture. Where things sit on the sheet means nothing; only the connections matter. The PCB editor reads those connections and turns them into the copper tracks you will route later.

Symbol vs footprint

- **Symbol:** the drawing on the schematic, a box with numbered pins. It says *what connects*.
- **Footprint:** the copper pads and holes on the PCB. It says *where the physical part's legs land*.
- They are linked **by pin number**. Symbol pin 1 goes to footprint pad 1. That is why the SK6812 symbol and the SK6812MINI-E_fixed footprint must both use 1 VDD, 2 DOUT, 3 VSS, 4 DIN, and they do.
- The pin **names** printed on a symbol are only labels. Your XIAO symbol shows SAMD21 names like PA02_A0_D0, but pin 1 is still D0. Wire by D-number.

Net

A **net** is one electrical node: every pin that touches it is joined in copper. GND is one net, L0 is another. When you draw a schematic, you are really building a list of nets.

Three ways to join pins into a net

1. **Wire (W):** a green line from pin to pin. Best for short, local connections, like a switch to its own diode.
2. **Net label (L):** a name stuck on a wire end. Every label with the **exact same text** is the same net, even across the sheet. Use labels for anything that travels far (L0, M1, SDA). They keep the sheet readable.
3. **Power symbol (P):** a special global label such as +5V, +3V3 or GND. Every +5V symbol is the same net.

A wire end must land exactly on a pin's tip or on another wire. A loose end shows a small circle or square, which means it is not connected. Where three or more wires meet, KiCad adds a **junction dot**; no dot means the wires only cross.

Power symbols and PWR_FLAG

Power symbols only **name** a net. They do not create power. Your board has exactly one real supply, the USB cable, and the XIAO turns it into two outputs:

- **+5V**, straight from USB, for the 12 LEDs.
- **+3V3**, made by the regulator on the XIAO, for the OLED.

KiCad cannot see the USB cable, so ERC would complain that nothing drives +5V or GND. A **PWR_FLAG** tells ERC that the net is powered. Put **exactly one** on each rail: +5V, +3V3 and GND. You have three on GND, so delete two and move one to +3V3.

No-connect flag (Q)

A small X on a pin you are deliberately leaving empty, such as the last LED's DOUT. Without it, ERC reports the pin as forgotten.

Reference and value

- **Reference** (SW1, D5, U1, LED3): the unique name of one part. Two parts must never share one. Right now your encoder and your first key are both SW1.
- **Value** (1N4148, OLED_0.91): what the part is. It is only text, for people.
- **Annotation** (Tools, Annotate Schematic) numbers all references automatically.

ERC (Electrical Rules Checker)

Inspect, Electrical Rules Checker. It finds unconnected pins, labels nobody else uses, unpowered nets and duplicate references. It cannot know your design is *correct*: a backwards diode passes ERC. **Run it after every block and aim for 0 errors.**

Crash course 2: the circuits on TechPad

The XIAO has exactly 11 usable pins (D0 to D10), and TechPad uses all of them. Every circuit below is designed around that limit.

Pin plan (official, from feasibility.md)

XIAO pin	Net label	Goes to
D0	ENC_A	Encoder A
D1	ENC_B	Encoder B
D2	L0	Matrix line L0
D3	L1	Matrix line L1
D4	SDA	OLED SDA
D5	SCL	OLED SCL
D6	L2	Matrix line L2
D7	LED_DIN	LED1 DIN
D8	M0	Matrix line M0
D9	M1	Matrix line M1

XIAO pin	Net label	Goes to
D10	M2	Matrix line M2
5V	+5V	All 12 LED VDD
3V3	+3V3	OLED VCC
GND	GND	Everything

The D-pins are **not 5 V tolerant**. Nothing that can carry 5 V may touch them.

Diodes: one-way valves

A diode lets current flow from the **anode** to the **cathode** only. The cathode is the end with the **bar**, on the symbol and on the real 1N4148. In a key matrix, the diode stops current from sneaking backwards through other pressed keys. Without diodes, pressing three keys in an L shape makes a fourth key look pressed (called ghosting).

A normal key matrix (what you drew)

Keys sit at the crossings of rows and columns. The firmware pulls one row LOW at a time and checks which columns read LOW. 3 rows times 3 columns gives 9 keys on 6 pins. That is exactly your current 3x3, and it is correct as far as it goes. But TechPad needs 13 switches, and you have no spare pins.

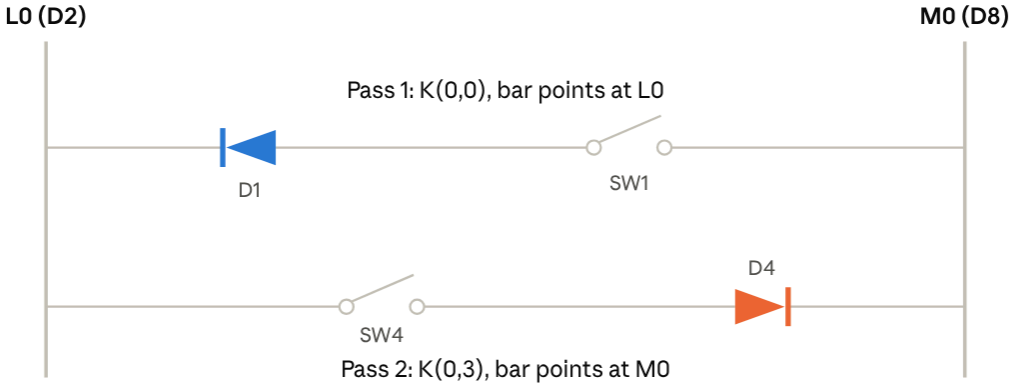
The duplex matrix (what TechPad needs)

The trick is to put **two switches at a crossing**, with their diodes pointing **opposite ways**. The two line groups are called L (L0 to L2) and M (M0 to M2). The firmware scans twice:

- **Pass 1:** drives each L line LOW and reads the M lines. It sees switches whose diode bar points at **L**.
- **Pass 2:** drives each M line LOW and reads the L lines. It sees switches whose diode bar points at **M**.

Each crossing can hold 2 switches, so 9 crossings give 18 positions. TechPad uses 13 of them.

Two switches share one crossing; the diode direction picks the pass



one duplex crossing, L0 x M0 · 2 switches, 2 diodes

Both switches connect the same two wires. Pass 1 sees SW1 because its diode bar faces L0. Pass 2 sees SW4 because its bar faces M0.

Crossing	Pass 1 (bar toward L)	Pass 2 (bar toward M)
L0 x M0	K(0,0)	K(0,3)
L0 x M1	K(0,1)	Encoder push
L0 x M2	K(0,2)	spare
L1 x M0	K(1,0)	K(1,3)
L1 x M1	K(1,1)	spare
L1 x M2	K(1,2)	spare
L2 x M0	K(2,0)	K(2,3)
L2 x M1	K(2,1)	spare
L2 x M2	K(2,2)	spare

K(row, column): row 0 is the top row and column 0 is the left. Columns 0 to 2 form an ordinary 3x3 matrix (Pass 1). Column 3 and the encoder push ride along in Pass 2. **Self-check: 13 diodes, with 9 bars on an L net and 4 on an M net.**

Rotary encoder (EC11)

Turning the knob makes pins A and B click to GND in a pattern, and the order tells the firmware the direction. Common pin C goes to GND. A and B go straight to D0 and D1, with no resistors, because the RP2040 has internal pull-ups. The push-click is a separate switch on pins S1 and S2, and it joins the matrix as switch 13.

OLED (I2C)

I2C uses 2 wires: SDA (data) and SCL (clock). The module already has pull-up resistors on board, which pull SDA and SCL up to its VCC. That is why VCC must be **+3V3**: on 5V, those pull-ups would push 5V into D4 and D5. The pin order on your module is GND, VCC, SCL, SDA. Check it against the printing on the real board before you trust it.

SK6812 MINI-E LED chain

Each LED reads colour data on DIN, keeps its own part, and passes the rest out on DOUT. So the LEDs form one daisy chain: D7 to LED1 DIN, LED1 DOUT to LED2 DIN, and so on to LED12. Each LED also needs +5V on VDD and GND on VSS. Snake the order through the keys (row 0 left to right, row 1 right to left, row 2 left to right) so the data tracks stay short. Twelve LEDs at full white draw about 720 mA, more than USB's 500 mA, so the firmware will cap brightness at about 40%.

KiCad basics

Press these keys with the mouse over the sheet. Most act on whatever is under the cursor.

Key	Does
A	Add a symbol (opens the chooser; type to search)
P	Add a power symbol (+5V, +3V3, GND, PWR_FLAG)
W	Draw a wire; click to bend, double-click to end
L	Add a net label
Q	Add a no-connect flag
T	Add a text note
M	Move (wires stay put, so connections can break)
G	Drag (wires follow; use this to tidy)
R	Rotate
X / Y	Mirror
E	Edit properties (reference, value, footprint)
Ctrl+D	Duplicate the selection
Delete	Delete

Key	Does
Ctrl+S	Save

A good working habit: build one block at a time, then save and run ERC. Keep a 50 mil (1.27 mm) grid on (View, Grid) so pins and wires snap together. If a part refuses to connect, the grid is usually off.

Library setup (fix this first)

Your sym-lib-table and fp-lib-table point at B:/Aarav/... . That path works only on your PC, so anyone cloning the repo, including Hack Club reviewers, sees "missing library".

1. Preferences, Manage Symbol Libraries, **Project Specific Libraries** tab. Change the two paths to:
   - `${KIPRJMOD}/../assets/kicad_care_package/XIAO_RP2040.kicad_sym`
   - `${KIPRJMOD}/../assets/kicad_care_package/SK6812MINI-E.kicad_sym`
2. Preferences, Manage Footprint Libraries, Project Specific Libraries tab. Change the path to `${KIPRJMOD}/../assets/kicad_care_package/footprints.pretty` and rename the nickname from `footprints` to `hackpad`.
3. **Keep** the symbol nicknames (XIAO_RP2040, SK6812MINI-E). Your placed parts are linked to those names, and renaming them would break the links.
4. Check that neither table shows a red or yellow warning icon.

`${KIPRJMOD}` means "the folder this .kicad_pro file is in", and `..` goes up one level to My_TechPad. The guide also says the project folder should be `PCB` in capitals, which matters on GitHub. Close KiCad before you rename the folder.

Build it step by step

Nine steps take your current file to a finished schematic. After each one, save and run ERC. Ignore "unconnected" errors for blocks you haven't built yet, and fix everything else before moving on.

Step 1: Power

1. Keep the +5V symbol on XIAO pin 14 (5V) and on every LED VDD.
2. Press P, place **+3V3** on XIAO pin 12 (3V3).
3. Keep GND on XIAO pin 13.
4. PWR_FLAGs: delete two of the three on GND, then wire the third one to +3V3. You should end with one each on +5V, +3V3 and GND.

Check: ERC shows no "power input not driven" error.

Step 2: Labels on the XIAO

From each D-pin, draw a short wire stub (W), then press L and put the label on the stub's end. Use the pin plan table in Crash course 2. Spell every label exactly: `L0` with a zero, not the letter O.

Check: 11 labels on 11 pins, and each label's little marker turns solid when it touches the wire.

Step 3: Rename your matrix labels

Your 9 branches already run column label, then switch, then diode, then row label, with the bar toward the row. That is exactly a Pass 1 branch, so rename the labels (double-click each one):

Old	New
R1	L0
R2	L1
R3	L2
C1	M0
C2	M1
C3	M2

Your switches are numbered **down the columns** (SW1, SW2, SW3 all sit on C1). The wiring is fine; you will fix the numbers in step 8.

Step 4: Add the 4 reversed (Pass 2) branches

These run the **other way**: L label, then switch, then diode anode, then diode cathode (bar), then M label.

Branch	Wiring	Key
1	L0, switch, diode, M0	K(0,3)
2	L1, switch, diode, M0	K(1,3)
3	L2, switch, diode, M0	K(2,3)

Branch	Wiring	Key
4	encoder S2, diode, M1 (and encoder S1 to L0)	encoder push

Shortcut: select one existing branch, press Ctrl+D, then swap its two labels. The diode keeps its direction relative to the switch, so the bar lands on the M side.

Check: count 13 diodes, with 9 bars touching an L label and 4 touching an M label.

Step 5: Encoder turning

Encoder A to label `ENC_A`, B to label `ENC_B`. C is already on GND, which is correct. S1 and S2 were done in step 4.

Step 6: OLED

On the Conn_01x04 you placed: pin 1 to GND, pin 2 to **+3V3**, pin 3 to label `SCL`, pin 4 to label `SDA`. Set its Value to `OLED_0.91_SSD1306`.

Step 7: LED chain (12 LEDs)

1. Press E on one LED and change its Reference from D12 to **LED1**. The LED symbol defaults to D, which would mix the LEDs up with the diodes.
2. Duplicate until you have 12. Each one: VDD (pin 1) to +5V, VSS (pin 3) to GND.
3. Data chain: `LED_DIN` label to LED1 DIN (pin 4). LED1 DOUT (pin 2) to LED2 DIN, and so on to LED12.
4. LED12 DOUT: press Q for a no-connect flag.
5. Press T and add the note: "LED order: row 0 L to R, row 1 R to L, row 2 L to R".

Step 8: Mounting holes and numbering

1. Place 4 MountingHole symbols, with no wires.
2. Renumber so the references follow the keys, left to right and top to bottom: SW1 = K(0,0), SW2 = K(0,1), SW3 = K(0,2), SW4 = K(0,3), up to SW12 = K(2,3). The encoder becomes SW13. Give each diode its switch's number (D1 next to SW1), and the encoder push diode is D13.
3. Do it by hand with E, or run Tools, Annotate Schematic with "Reset existing annotations" and then fix the few that come out wrong.

Step 9: Final ERC and screenshot

Run ERC until it shows **0 errors**, and read the warnings too. Then take a screenshot or use File, Plot, PDF for your README, since Hackpad requires a schematic image.

Final check before review

Tick these off, then send the file to Claude in the project thread for a netlist check before you start footprints and the PCB.

- [ ] Library paths use ${KIPRJMOD}, and no table shows a warning icon
- [ ] All 11 XIAO D-pins carry a label that matches the pin plan
- [ ] +5V on XIAO 5V and all 12 LED VDD; +3V3 on XIAO 3V3 and OLED VCC only
- [ ] Exactly 3 PWR_FLAGs: one each on +5V, +3V3, GND
- [ ] 13 switches: SW1 to SW12 in key order, plus the encoder as SW13
- [ ] 13 diodes: 9 bars toward L, 4 bars toward M
- [ ] Encoder: A to ENC_A, B to ENC_B, C to GND, S1 to L0, S2 to D13 to M1
- [ ] OLED: 1 GND, 2 +3V3, 3 SCL, 4 SDA
- [ ] 12 LEDs named LED1 to LED12, chained from LED_DIN, no-connect on LED12 DOUT
- [ ] 4 mounting holes
- [ ] No duplicate references
- [ ] ERC shows 0 errors
- [ ] Schematic screenshot saved for the README

ERC messages and their fixes

ERC says	Usually means	Fix
Pin not connected	A wire end misses the pin tip	Drag the wire onto the pin, or add Q if it is meant to be empty
Input power pin not driven	No PWR_FLAG on that rail	Add one PWR_FLAG to the net
Label not connected anywhere else	A typo, or the label is on only one end	Check spelling (L0 vs LO), then add the matching label
Duplicate reference	Two parts share a name (like SW1)	Annotate, or edit one by hand
Pin connected to other pins, but no driver	Usually a power net with no flag	Add PWR_FLAG

Mistakes ERC cannot catch

- **A backwards diode.** It passes ERC, but the key moves to the wrong pass. Count 9 bars on L and 4 on M.

- **OLED VCC on +5V.** It passes ERC, but it can damage the XIAO. VCC must be +3V3.
- **Wiring by pin name instead of D-number** on the XIAO. Pin 8 is D7, even though the symbol says PB09_A7_D7_RX.
- **Wires that cross without a junction dot.** They are not connected. Use labels instead of long crossing wires.